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NAME VARUN CHAUHAN

ROLL NO. 2401010276

SUBJECT DBMS

ASSIGNMENT 2

SEMESTER IVth

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'Section - A'

Ques - ① Fill in the blanks -

(a) ER diagram
(B) Relational model
(c) Object - oriented model
(d) Integrity constraints

'Section - B'

Ques - ② In a healthcare systems, discuss how data abstraction can simplify interaction between healthcare professionals and patient records.

Data abstraction means hiding complex database details and showing only necessary information to users.

In a healthcare system, data abstraction helps doctors, nurses and administrators access patient information easily without dealing with complex database structures.

It works in 3 levels -

(a) Physical level -
This level defines how data is actually stored in the database using storage structures. Healthcare professionals don't interact with this level.

(B) Logical level -
This level describes what data is stored in the database and the relationship b/w the data.
For example - relationships between patients, doctors, medical records and treatments.

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(c) View level -
This level shows only the required information to specific users. For ex.
- Doctor can see diagnosis and treatment history.
- Nurses can view medication schedules.

Thus, data abstraction makes it easy for healthcare staff can easily retrieve patient records, update information.

Ques-③ Explore the benefits and the challenges of implementing data abstraction in healthcare systems.

Benefits -

(a) Easy to use - Healthcare professionals can access information without technical database knowledge

(B) Improved data security - Sensitive patient data is protected by restricting access.

(c) Data consistency - Abstraction ensures that all users view consistent and accurate information.

(d) Better system maintenance - Database changes can be made internally without affecting users.

(e) Improved efficiency in healthcare operations -
Quick access to patient records helps doctor make faster decisions.

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challenges -

(a) implementation complexity
(b) performance overhead
(c) high development cost
(d) security risks if poorly implemented
(e) training requirements

Despite these challenges, data abstraction significantly improves the efficiency and reliability of healthcare data management.

'Section-C'

Design a database for a university incorporating entities like students, courses and professors. Using the ER model, illustrate the relationships between these entities. Include attributes for each entity to capture relevant info. such as student names, course details, and professor info.

- student attributes -
  (student_ID (PK), Name, Email, Phone, Dept.)

- course attributes -
  (course_ID (PK), Name, credits, Dept)

- Professor attributes -
  (PROFF_ID (PK), Name, Dept)

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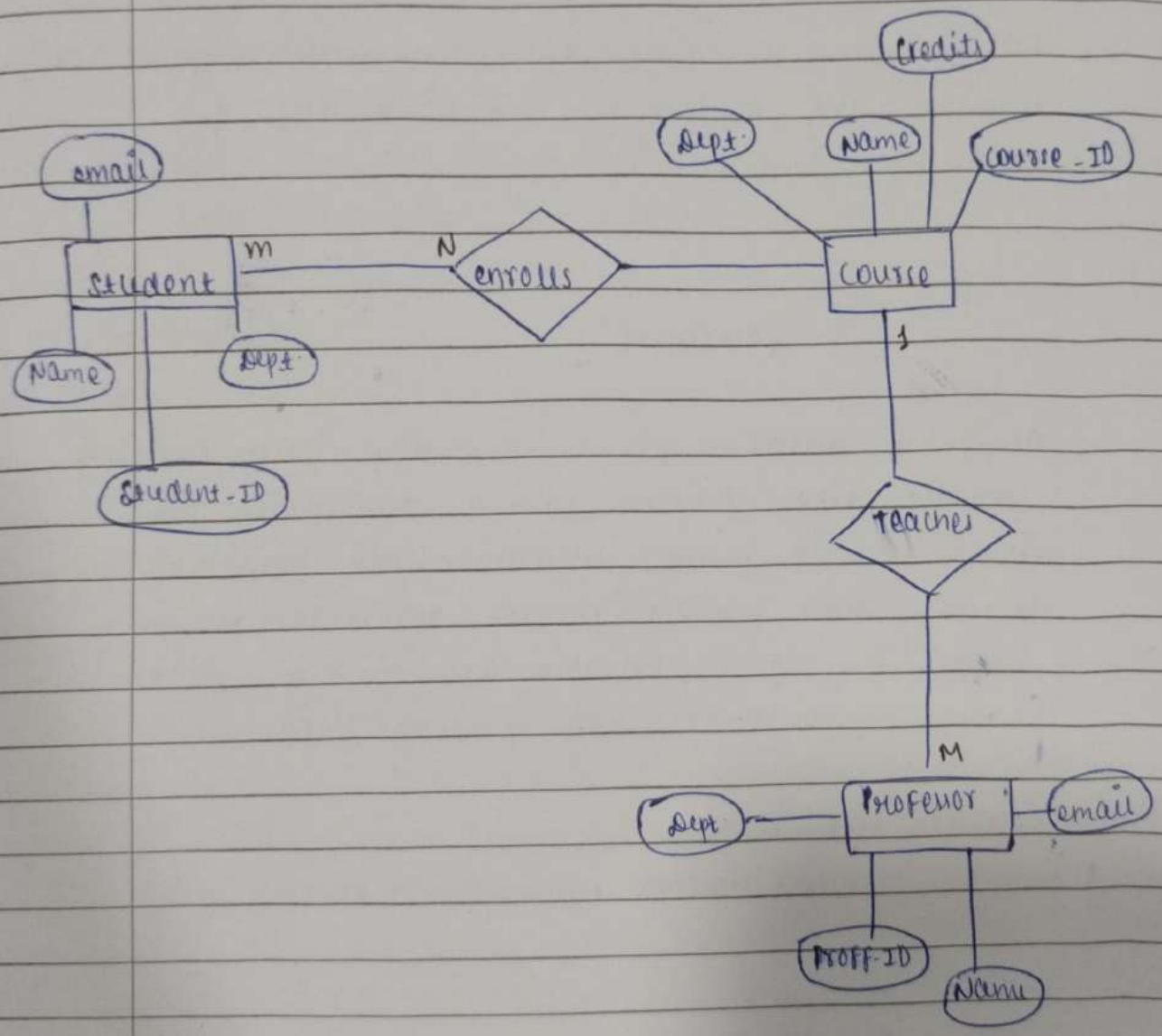
Relationships -

(a) Enrollment - (student - course)
(b) Teaching - (Proff - ~~course~~ course)

Diagram:



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